

Bala Kishore Penna

Full Stack Developer

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Profile

Experienced **Full Stack Developer** with hands-on experience in **Smart Home, IoT, and SmartMonitoring** solutions. Proficient in **JavaScript, React, Node.js, MongoDB, Express, AWS, Redis, NGINX, and MQTT**. Skilled in developing **Alexa Smart Home Skills** and certifying them for third-party brands. Adept at building and optimizing scalable **MERN stack applications**, with a strong focus on performance tuning, real-time data processing, and efficient data workflows.

PROFESSIONAL EXPERIENCE

Full Stack Developer

Quest Global – (Client: Reliance Jio, Mumbai, Hyderabad)

March 2023 – Present

Team: Automation and Surveillance (IoT Devices)

Contributing to web solutions for IoT device monitoring, control, and analytics with real-time data streaming. Supporting both monolithic and **microservice** architectures to deliver scalable solutions for **B2B** and **B2C** markets (hotels, temples, households).

Key Projects: Jio SelfCare, Brands portals, Jio Smart Living, voice skills Alexa.

Responsibilities:

- Developed RESTful APIs using **Node.js** and **Express.js**, improving system efficiency by **40%**. Optimized SQL/NoSQL database queries, boosting API performance and reducing load. Applied Redis caching to cut request latency by **50%**.
- Managed deployments using Linux commands and **PM2**.
- Reduced API response time from **800ms** to **300ms** using caching and query optimization.
- Built user-friendly UIs for device control, analytics, and monitoring.
- Developed intuitive UIs for IoT solutions, including device control, analytics, event monitoring, and gateway interactions, enhancing user engagement.
- Investigated and resolved critical production issues by analyzing logs, **debugging** code, and collaborating with cross-functional teams, ensuring minimal downtime and uninterrupted user experience.
- Integrated **Alexa Smart Home Skills** with **IoT** products to enable seamless voice control.
- Handled all Alexa-related activities including debugging, maintenance, integrations, development of new skills, and partnerships with third-party brands to expand the ecosystem and ensure certification compliance.
- Built REST APIs with **Python(Flask/FastAPI)** using MongoDB and Redis for fast data access.

KEY SKILLS

Development Technologies: React.js, Node.js, Express.js, MongoDB, Redis, MQTT, Rabbit MQ , EMQX, CSS, Nginx, AWS lambda

Programming Languages: JavaScript, python

Libraries/Frameworks: JWT, Bootstrap, Tailwind CSS, Ant Design.

Tools: Git, Figma, Postman, VS Code, Swagger, Gerrit, Redmine, AWS lambda and cloud watch, **Alexa** developer console.

Education

Bachelor of technology

- Amrita sai institute of science and technology - **May 2022**

Activities and Interests

- environmental conservation, art, hiking, cooking, travel, fitness

ACHIEVEMENTS

- Awarded as standout performer of the Year 2024 for outstanding performance and contributions on JioSmartLiving products.